

Presentation on Solution of Rapid EV Charging

DATE - 23/09/2026

By Nikhil Kushwah

EE, 7th Sem, SATI Vidisha

E-Mail : nikhilkushwah078@gmail.com

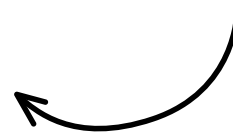
Mo. NO. : 7987260228



We already discuss the problem



Watch this video on
Youtube to
understand the
problem

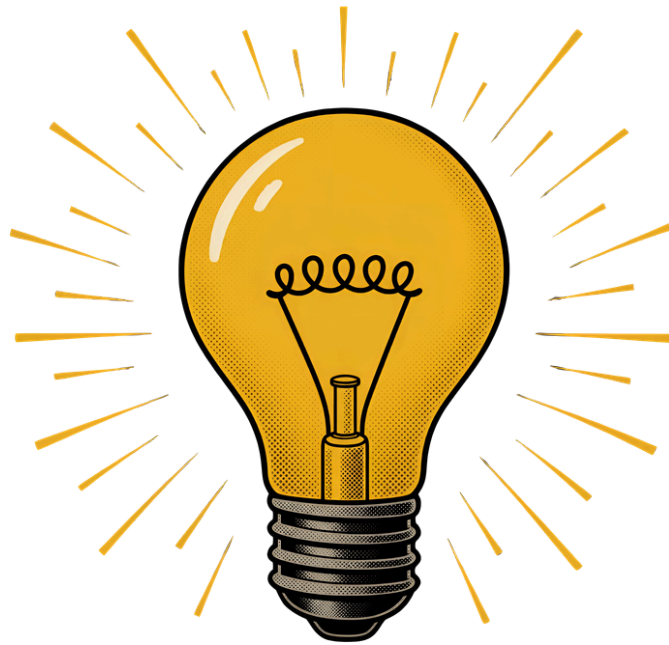


https://youtu.be/_GY06RAAtI1g?si=pScHVihUnqALHQmG

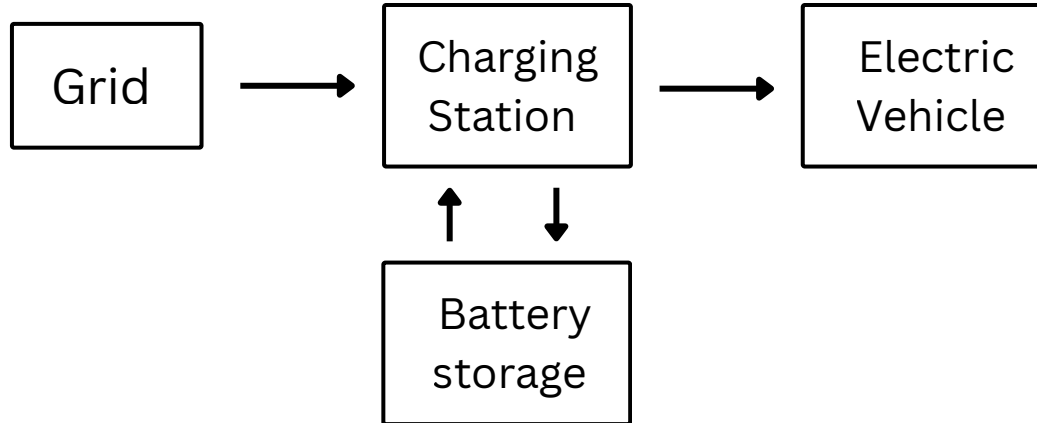
Overview -

- Rapid EV charging need very amount of power in very short duration of time like such as 350KW per car at a time
- Which create an sudden load on gird when we talk about multiple car charging at a time
- Grid cannot handle that much of variable load

Lets talk about
the solutions



Battery based buffer system at charging station -



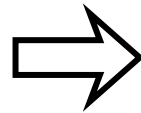
Elon Musk says Tesla's energy storage business "very strong", actual forecast for 2026 is weak

New information out of Tesla's Q1 2026 earnings call overnight shows the energy storage business growth may have stalled.

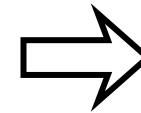


By Tristan Rayner | Apr 23, 2026

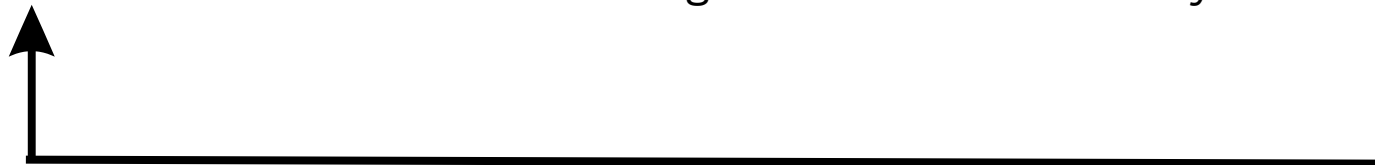
Charging station first charge its battery storage system slowly



Then it rapid charge the EV without creating the sudden load on grid



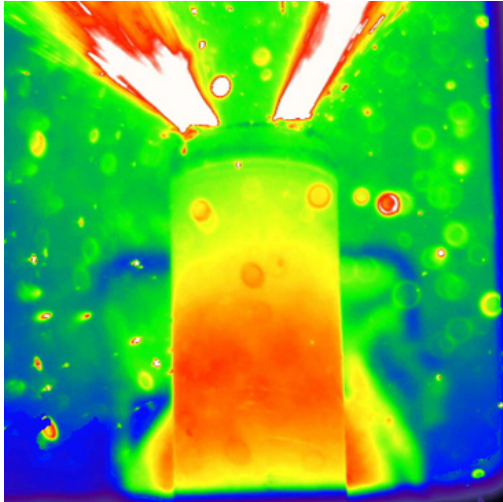
Then after discharging again start charging the battery storage system



Cycle goes on

Problems with the battery storage -

- Need of multiple batteries to full fill the demands (High initial investment)
- Large physical space need for batteries
- Battery degradation due to continues charging and discharging
- Round trip efficiency may cause problem
- Li-ion battery has very high Safety and fire risk
- Thermal management is very difficult



Working physics Li-ion Battery

$$\Delta G = -nFE$$

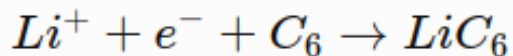
$$E = -\frac{\Delta G}{nF}$$

where:

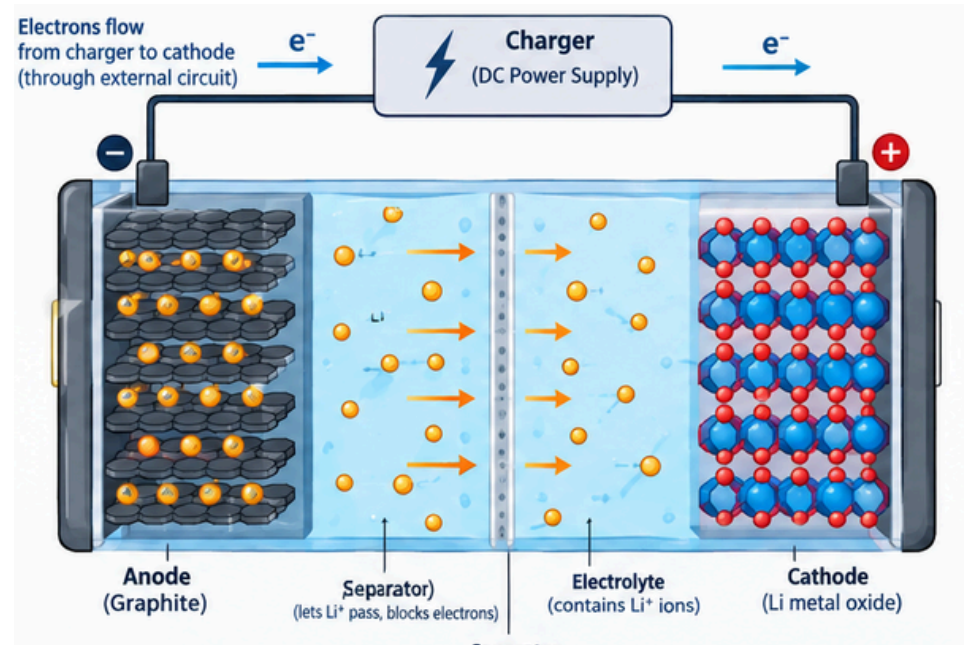
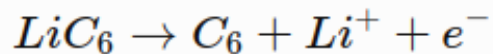
- E = cell voltage
- ΔG = Gibbs free-energy change
- n = number of electrons transferred
- F = Faraday constant

$$F \approx 96485 \text{ C/mol}$$

Charging



Discharging



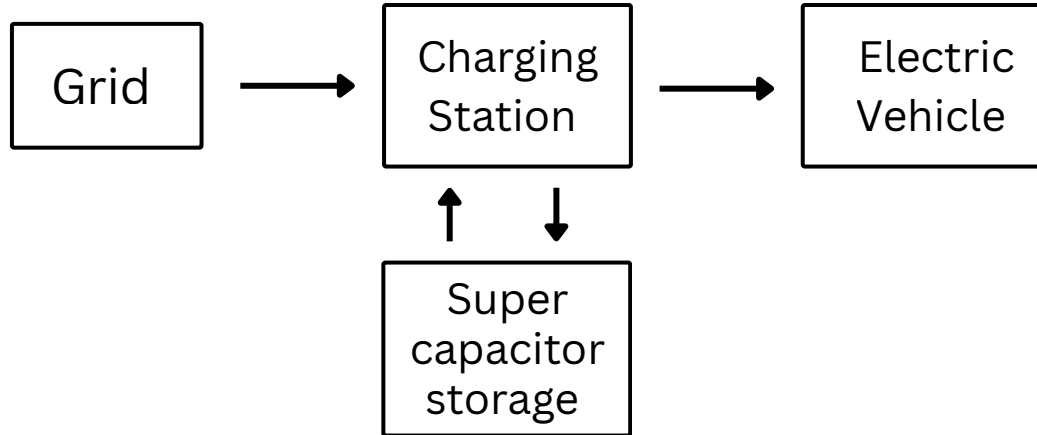
Involve physically
electron exchange

Which limit the flow
of electron and
increase the internal
resistance

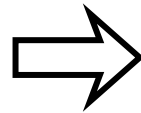
Battery will degrade after
threshold cycles

Also create heat due
to flow of electron
during charging and
discharging

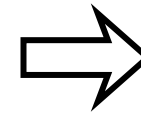
Super-Capacitor Based Energy Storage Systems -



Charging station first charge its super-capacitor storage system slowly



Then it rapid charge the EV without creating the sudden load on grid



Then after discharging again start charging the super capacitor storage system



Cycle goes on

Working physics

Super-capacitor

$$E = \frac{1}{2}CV^2 \quad u = \frac{1}{2}\epsilon E^2$$

A supercapacitor isn't simply a normal capacitor with two metal plates.

It uses:

- porous electrodes
- electrolyte
- ions
- enormous internal surface area

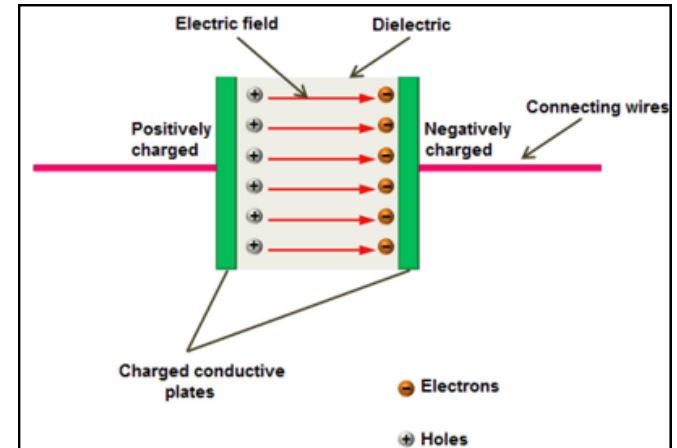
When voltage is applied:

- positive ions move toward the negative electrode
- negative ions move toward the positive electrode

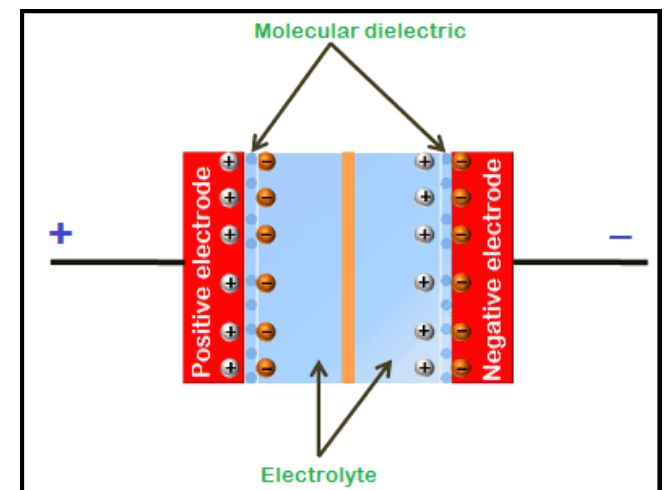
They form extremely thin **electric double layers** at the interfaces.

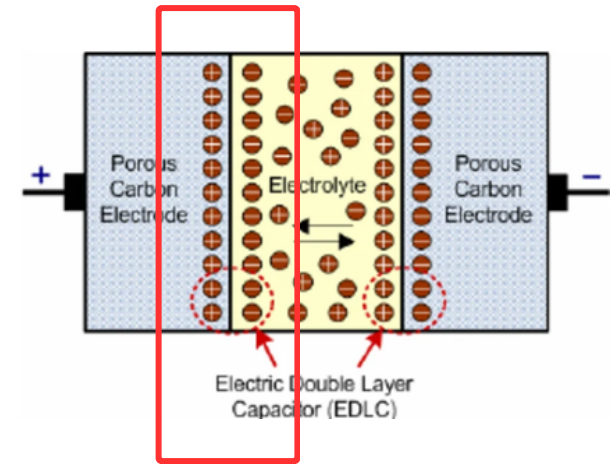
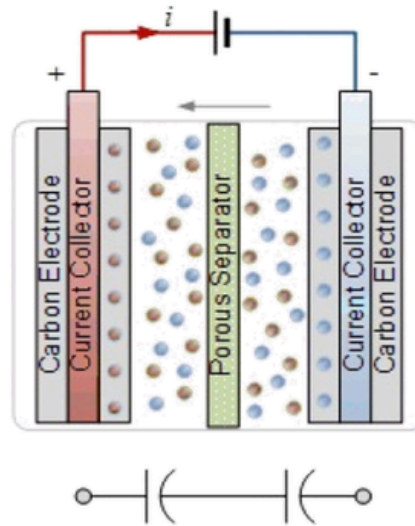
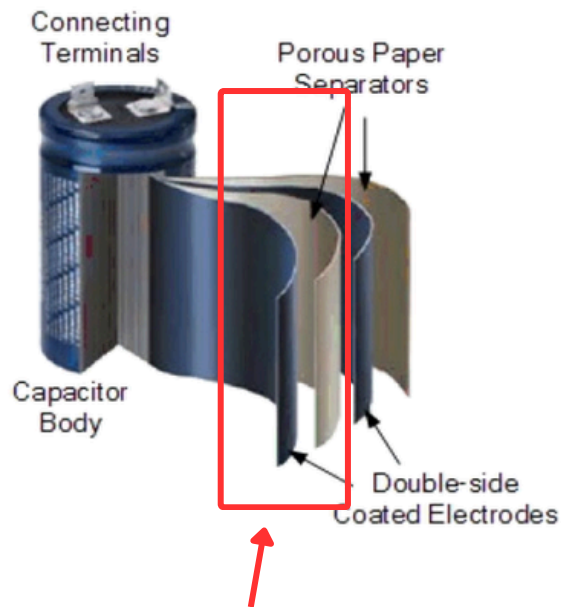
So the effective capacitor has an extremely small charge-separation distance.

Normal capacitor



Super capacitor





Because of electric double layer the charge separation is very low (Distance between two plates) the capacitance is very high that is why

Supercapacitor have enormous capacitance

$$C = \frac{\epsilon A}{d}$$

$C \uparrow$ if $A \uparrow$

$C \uparrow$ if $d \downarrow$

Then form the formula -

$$E = \frac{1}{2} CV^2$$

We can determine the energy stored in the capacitor

The voltage of the capacitor depends on there physical construction that is why we in charge the capacitor form 100v to 1000v without any major limitation

Example -

Suppose we have a capacitor:

- Capacitance: **C = 100 F**
- Maximum voltage: **V = 48 V**
- ESR: **R = 10 mΩ = 0.01 Ω**

How much energy does it store?

$$E = \frac{1}{2}CV^2$$

$$E = \frac{1}{2}(100)(48)^2$$

$$E = 115,200 \text{ J}$$

$$E = \frac{115200}{3600} = \boxed{32 \text{ Wh}}$$

So this 100 F, 48 V capacitor bank stores only 32 Wh.

But how much power can it deliver?

$$P = VI$$

Suppose the capacitor can deliver 1000 A for a short period.

$$P = (48)(1000)$$

$$\boxed{P = 48,000 \text{ W} = 48 \text{ kW}}$$

So we have:

Energy = only 32 Wh

but

Power = 48 kW

That's the key point.

How long could it theoretically deliver 48 kW?

$$t = \frac{E}{P}$$

$$t = \frac{32}{48} = 0.667 \text{ h}$$

That looks like 40 minutes, but this is misleading because the capacitor voltage cannot remain at 48 V while it discharges.

$$V = \frac{Q}{C}$$

When the current flow the charge reduces which decreases the voltage

Suppose we only use the capacitor from 48 V down to 24 V.

$$E = \frac{1}{2}C(V_{max}^2 - V_{min}^2)$$

$$E = \frac{1}{2}(100)(48^2 - 24^2)$$

$$E = 86,400 \text{ J}$$

$$E = \boxed{24 \text{ Wh}}$$

Now suppose it supplies 48 kW:

$$t = \frac{24}{48} = 0.5 \text{ h}$$

An even better example: 10-second EV power boost

Imagine an EV charger normally receives:

$$P_{grid} = 300 \text{ kW}$$

But the EV temporarily needs:

$$P_{EV} = 500 \text{ kW}$$

The supercapacitor supplies:

$$P_{SC} = 500 - 300 = \boxed{200 \text{ kW}}$$

for **10 seconds**.

Energy required:

$$E = Pt$$

$$E = (200,000)(10)$$

$$E = 2,000,000 \text{ J}$$

or:

$$E = \frac{2,000,000}{3600}$$

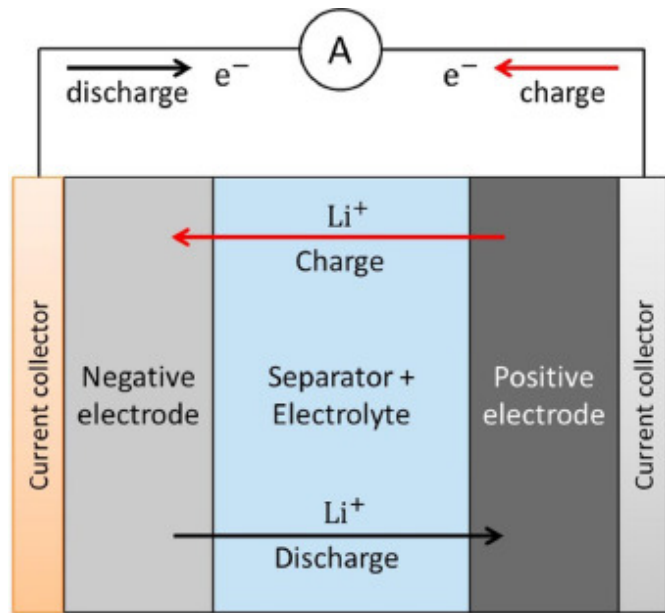
$$\boxed{E = 556 \text{ Wh}}$$

So the supercapacitor only needs to provide about **0.56 kWh**, but it must deliver:

$$\boxed{200 \text{ kW}}$$

This is exactly where a supercapacitor becomes attractive.

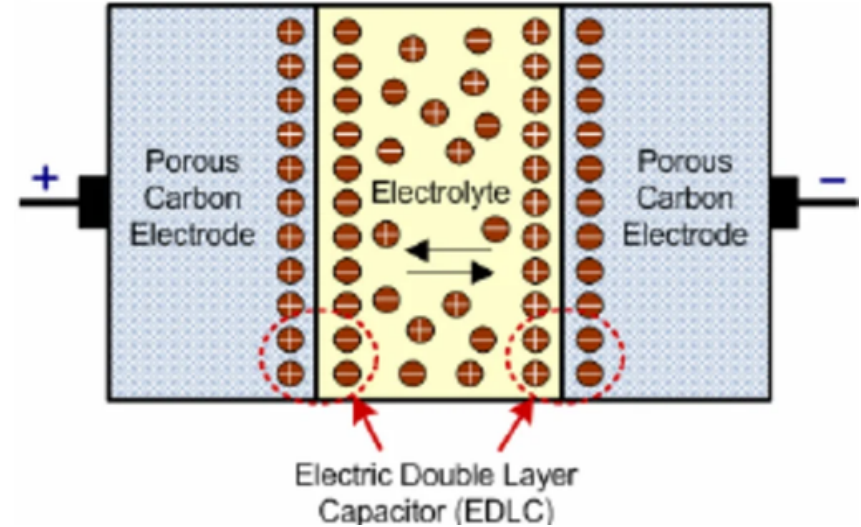
Comparison



Li-ion battery

- Lithium ions have to travel through the electrolyte and enter/leave electrode materials.
- electrolyte transport
- ion diffusion
- heat generation

Ion diffusion $\rightarrow$ chemical/electrochemical reaction $\rightarrow$ energy storage $\rightarrow$ slower process



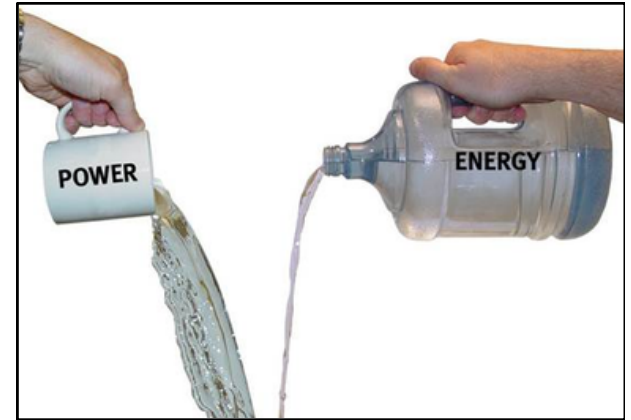
Super-Capacitor

- primarily moving charge onto surfaces without any actual ion exchange process
- Very fast ion movement
- Extremely large electrode surface area
- Low internal resistance
- Very high power density

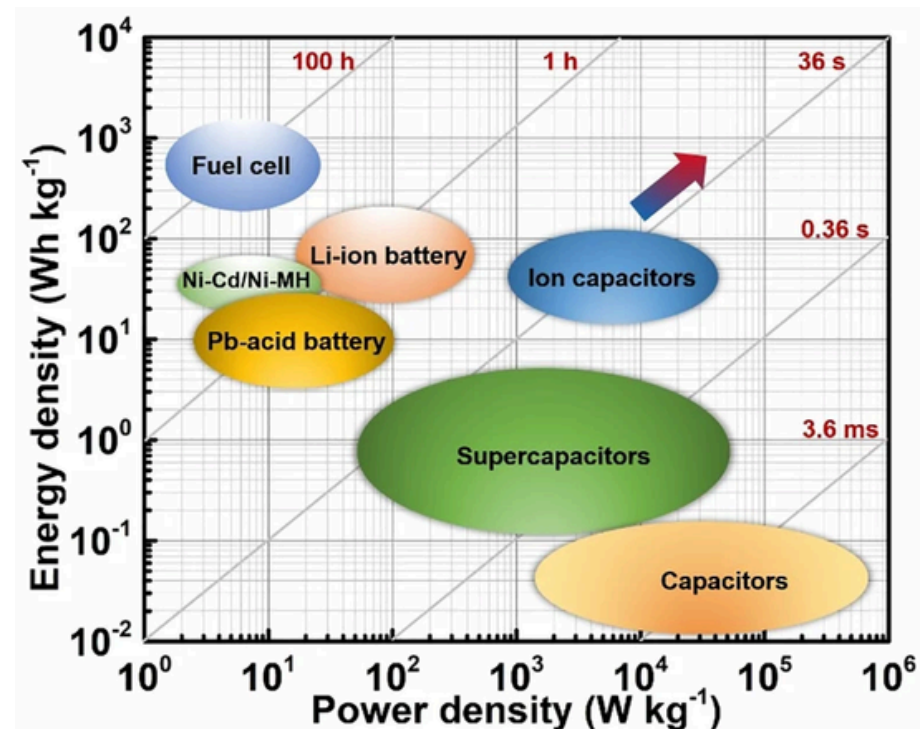
Ion movement $\rightarrow$ charge accumulation at electrode surface $\rightarrow$ electric-field energy storage $\rightarrow$ extremely fast process

Advantages of Super-capacitors over Li-ion battery system -

- Extremely high cycle life
- Very fast charging and discharging
- Very high power capability
- No degradation like lithium-ion battery degradation



Characteristic	Battery	Supercapacitor
Energy density	High	Low
Power density	Moderate	Very high
Charge/discharge speed	Moderate	Very fast
Cycle life	Limited	Very high
Response time	ms-seconds	milliseconds
Suitable for	Long-duration energy	Short-duration power peaks



Problems with the super capacitor based energy storage system -

- Ions still have to move through the electrolyte to form the electric double layer. At extremely high currents, ion transport can become a limiting factor.
- The electrolyte has finite ionic conductivity and resistance. At high current, this produces voltage drop and heat.
- Supercapacitors generally lose stored energy faster than Li-ion batteries when sitting unused
- This is the biggest limitation for your EV application. A supercapacitor can deliver enormous power but stores relatively little energy.
- Unlike a battery, its voltage changes significantly as it discharges:

Capacitance $\rightarrow C = \frac{Q}{\Delta V}$

Charge $\rightarrow Q$

Voltage $\rightarrow \Delta V$

permeability of the material between plates $\rightarrow \epsilon_r$

Area between parallel plates $\rightarrow A$

distance between plates $\rightarrow d$

dielectric constant $\epsilon_0 \approx 8.854 \times 10^{-12} \text{ F} \cdot \text{m}^{-1}$

$$C = \epsilon_r \epsilon_0 \frac{A}{d}$$

“Supercapacitors can replace Li-ion batteries for high-power buffering in rapid EV charging—bringing faster response, longer life, and higher power capability, with further improvements in energy density and system design.”

THANK YOU !